Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

A. It activates TLR5 on the surface of macrophages.

B. It primarily induces the Th1-cell response.

C. Peptidoglycan is its major antigen that induces an immune response.

D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

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- The patient 's symptoms and X -ray findings suggest community -ac quired pneumonia .

- The path ogen causing this condition is likely Stre pt oc occus pneumonia e , a common cause of community -ac quired pneumonia .

- Stre pt oc occus pneumonia e primarily induces the Th 1 -cell response , which is characteristic of bacterial infections .

- T LR 5 is more relevant for detecting bacterial flag ell in , and pe pt id og ly can is a component of the cell wall of Stre pt oc occus pneumonia e , but not the primary antigen for the immune response .

- Stre pt oc occus pneumonia e does not evade the immune response by encaps ulation ; it is encaps ulated itself and uses its capsule as a vir ulence factor .

Therefore , the most accurate description of the immune response elic ited by Stre pt oc occus pneumonia e would be the Th 1 -cell response .

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< answer >B . It primarily induces the Th 1 -cell response .</ answer >